



Built on Arbitrum Stylus

OMNIVERSE

Zero-Liquidation Prediction Markets with On-Chain Gaussian Defense

Rust + WASM math kernel · live on Arbitrum Sepolia

omniverse-99so.vercel.app

THE PROBLEM WITH

prediction-market LPs

Standard pm-AMM, at resolution

P(YES) 96%



LP capital exposed: 100%

LP inventory: all NO → ~\$0

Outcome: wiped out in one block

01

LP wipeout

When the event resolves, arbitrageurs grab the winning side in a single block. LPs are left holding the worthless token.

02

Full exposure

Every dollar of LP capital is tradeable at all times — even at 99% certainty, when the trade is a sure loss for the pool.

03

Liquidation risk

Borrowing against prediction positions means margin calls and cascading liquidations the moment price moves.

OUR INSIGHT: *liquidity is a risk surface*

01

Risk isn't uniform

Adverse selection explodes as price nears 0 or 1. The danger to LPs is a curve, not a constant.

02

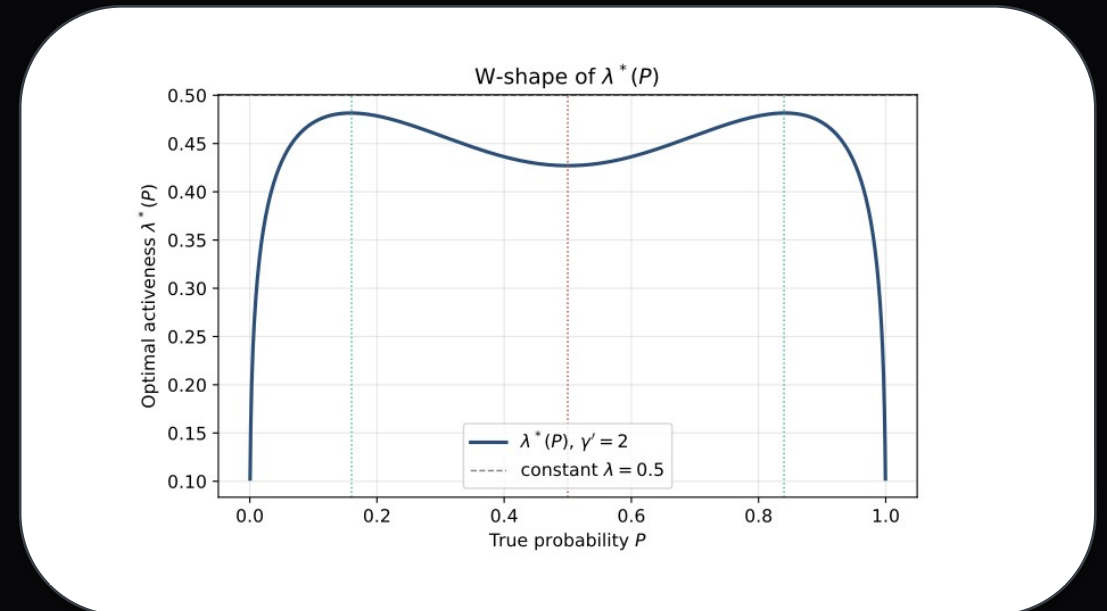
So shield it dynamically

The Gaussian λ^* curve pulls LP capital aside exactly when it is most exposed, and releases it near 50/50.

03

Match outcomes, not prices

Same-outcome collateral and debt cancel on resolution, so lending needs no liquidation engine at all.



From our whitepaper: $\lambda^*(P)$ is W-shaped — peaks at $P \approx 0.16/0.84$, $\rightarrow 0$ at the tails

RESEARCH TO *MVP*

01 pm-AMM — the Gaussian invariant (Paradigm)

$P = \Phi((y-x)/L)$. The closed-form prediction-market curve we build on. paradigm.xyz/2024/11/pm-amm

02 PA-AMM + Gaussian λ^* — our extension

We split reserves into active and passive each block, sized by $\lambda^*(\gamma', P)$, so LP exposure shrinks at the extremes. Original work for this build.

03 Arbitrum Stylus — the whole kernel in Rust/WASM

φ , Φ , Φ^{-1} , λ^* , and a Newton-Raphson swap solver, deployed and live on Sepolia at 18-decimal fixed point.

THE MATH *made simple*

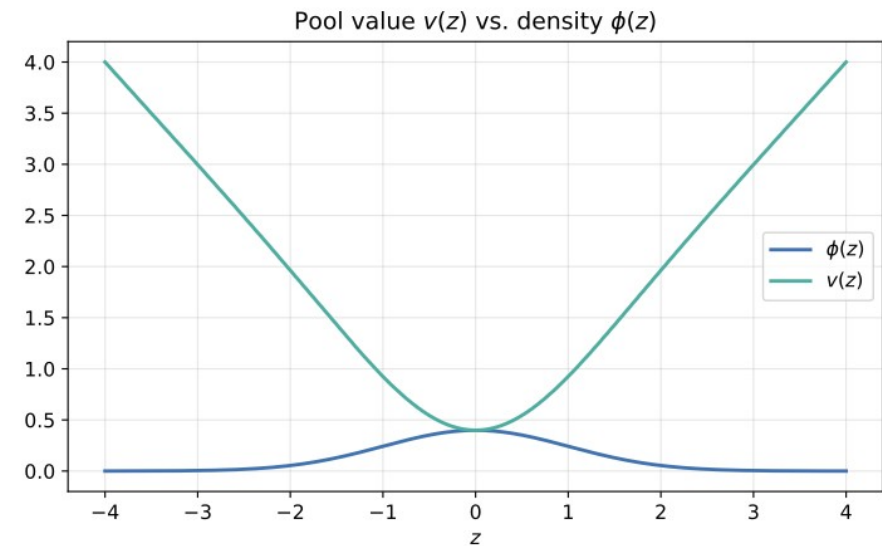
Marginal price = probability

$$P = \Phi((y - x) / L)$$

Defense grows near certainty

$$\lambda^*(P) \rightarrow 0 \text{ as } P \rightarrow 0 \text{ or } 1$$

Price is just the Gaussian CDF of the reserve imbalance. The further the market leans, the less LP money we leave in the line of fire. No oracle, no off-chain math.



Pool value $v(z)$ vs price-sensitivity $\phi(z)$ — the curvature that sets λ^*

WHY A W, *not a dome*

The obvious guess — most active at 50/50 — is provably wrong.

01 Activeness follows capacity

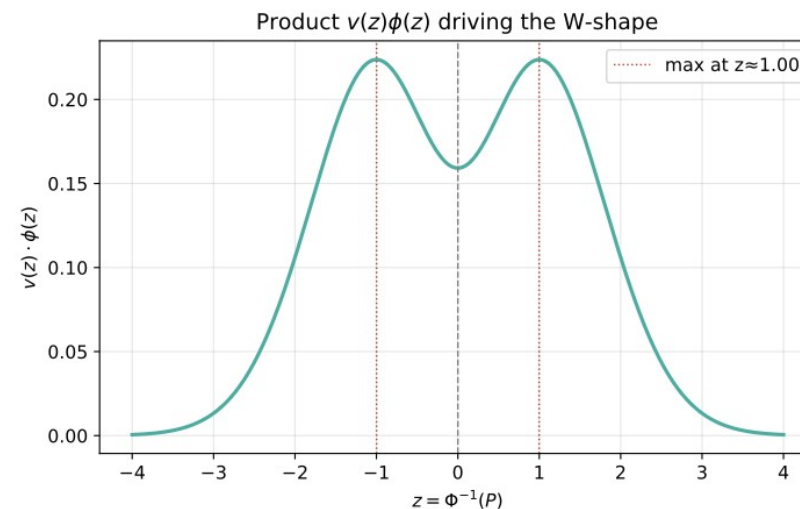
LPs should be most active where the pool's risk-adjusted capacity $v(z) \cdot \phi(z)$ is highest.

02 That capacity peaks off-center

$v(z) \cdot \phi(z)$ maxes at $z \approx \pm 1$ — $P \approx 0.16$ and 0.84 — not at 0.5 . (Lemma 1: $z=0$ is a local minimum.)

03 So λ^* is a W

It dips at 50/50, peaks off-center, collapses at the tails. A naive dome costs 2–3× more at the edges.



$v(z) \cdot \phi(z)$: twin peaks at $z \approx \pm 1$ are the origin of the W-shape

THREE-LAYER *defence*

Each layer bounds a different timescale of LP loss. Together: the most complete protection in the AMM literature.

1

Time-decaying liquidity

Pool depth shrinks as the event matures — bounds lifetime LVR, independent of volatility.

$$L_t = L_0 \cdot \sqrt{T - t}$$

2

Partially-active reserves

Only a λ -fraction trades each block; caps per-block adverse selection uniformly.

$$R_a = \lambda \cdot R_{\text{total}}$$

3

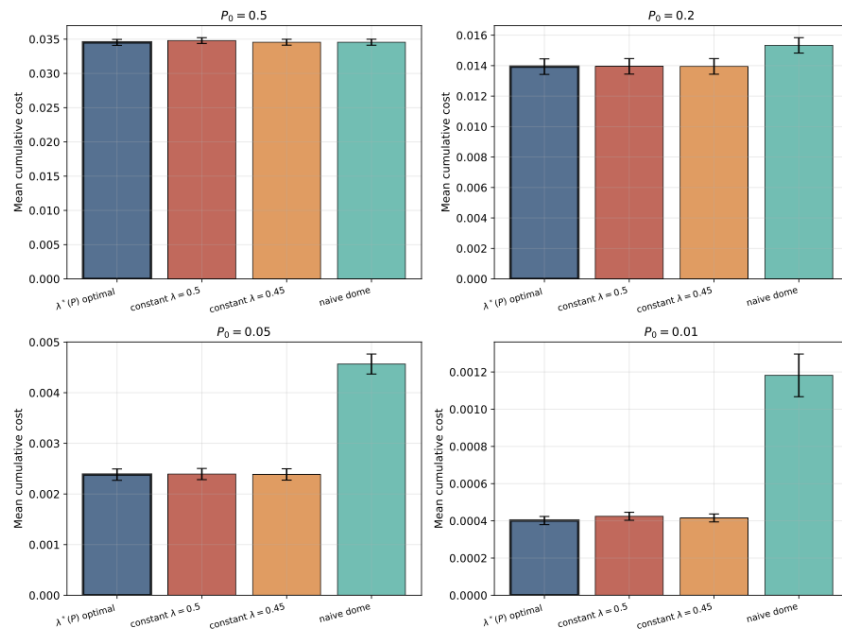
Gaussian $\lambda^*(P)$ · ours

Tightens the active fraction near resolution — automatically, no oracle, every block.

$$\lambda^*(P) \rightarrow 0 \text{ at the tails}$$

$$\text{Combined: } \ell_{\text{active}}(t, P) = \lambda^*(P) \cdot L_0 \cdot \sqrt{T - t}$$

PROVEN, *not hand-waved*



Monte Carlo: λ^* is the lowest-cost policy at every probability (200 paths)

01

Exactly AR(1)

Gap dynamics are exact at all orders — simulated / theoretical variance ratio 1.001 ± 0.028 .

02

Lowest cost, every regime

$\lambda^*(P)$ beats every constant policy and the dome across all 200 independent Monte-Carlo paths.

03

Up to 73% less exposure

Near resolution ($P = 0.999$) λ^* cuts the active LP fraction from 0.50 to 0.13 — the dome costs 2–3× more.

THE EVIDENCE *the data backs it*

Standard AMMs and oracle-based lending are measurably broken. Omniverse fixes each.

50%+

of AMM LPs lose money — arbitrage (LVR) beats fee revenue.

Fritsch & Canidio · ACM WWW 2024

→ Gaussian λ^* bounds lifetime LVR

\$150M+

of borrower value leaked to MEV bots through liquidations.

Pyth Network

→ outcome-matched lending: no liquidations

\$27M

of healthy positions liquidated on a single oracle glitch.

Aave · CoinDesk, Mar 2026

→ no oracle, so no false liquidation

\$21.5B

prediction-market volume — roughly 130× in one year.

Dune, 2025

→ the market we make LP-safe

WHAT WE *built*

01

On-chain Gaussian pricing engine

φ , Φ , Φ^{-1} and a Newton-Raphson swap solver in Rust, compiled to WASM, running via Arbitrum Stylus.

02

PA-AMM pool with dynamic λ^* shielding

Reserves split into active and passive every block; passive capital is hidden from informed flow.

03

ERC-1155 conditional tokens

Gnosis-style YES/NO outcome tokens with split, merge, and redeem against collateral escrow.

04

Zero-liquidation lending

Borrow same-outcome debt against same-outcome collateral; the position nets to zero on resolution.

05

Indexer + live dashboard

A Ponder GraphQL indexer and a React app, deployed end-to-end and reading state straight from chain.

ON-CHAIN IN *~5,000 gas*

The whole $\lambda^*(P)$ pipeline runs in a Rust/WASM kernel via Arbitrum Stylus.

Gas per block (once, not per swap)

$\Phi^{-1}(P) \cdot 50\text{-iter bisection}$ **~4,000**

$v(z), \gamma_G, \lambda^*$ **~1,000**

Total per block **~5,000**

$\approx 3\text{--}5\%$ overhead on a swap (100–200K gas). At 0.25s blocks, ~20K gas/sec — well within L2 limits.

Three ways to ship it

A

Full dynamic λ^* · recommended

Per-block $\lambda^*(P)$ straight from the kernel. Provably optimal, automatic tail protection.

B

Piecewise

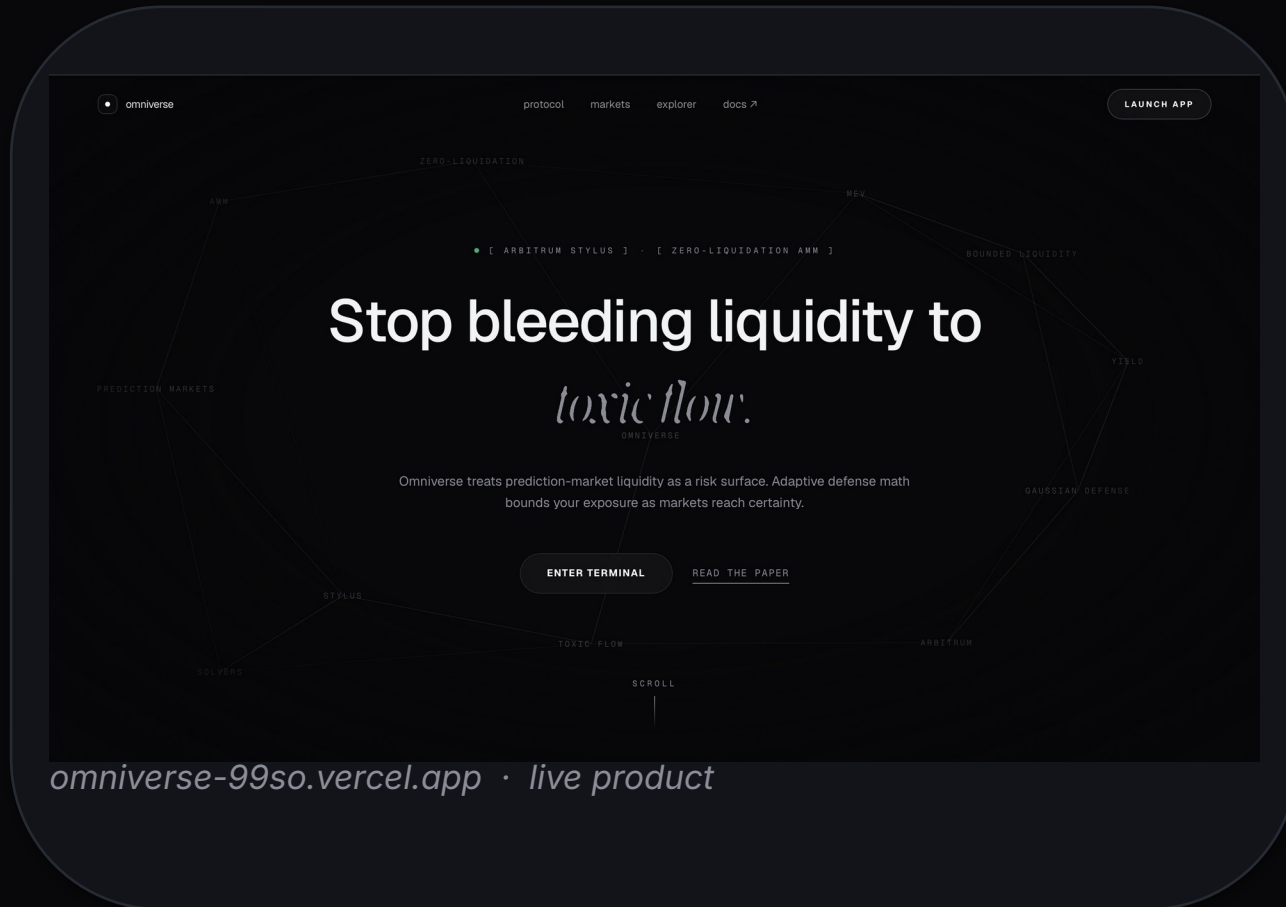
Dynamic λ^* only at the tails ($P < 0.05$ or > 0.95), constant ≈ 0.45 in between. Gas-free for 90% of states.

C

Constant + tail-freeze

Simplest to audit: a constant λ that halts trading in the extreme tail.

LIVE ON SEPOLIA *the attack*



0.50

seeded market, exactly fair

3 buys

escalating 2k / 4k / 6k attack

→ 0.89

probability climbs up the curve

63%

of LP capital pulled into the shield

FROM HACKATHON TO PRODUCT

what's next

01 Trustless resolution

Replace the owner-controlled resolver with a multi-agent AI oracle that can debate, vote, and abstain when unsure.

02 Smarter trading tools

Limit orders and 'basket' bets that let a trader express a whole distribution in a single transaction.

03 Consumer-friendly UX

Plain belief-meters instead of raw curves, push alerts, and one-tap betting.

04 AI agents as traders

A closed-form instant price is perfect for autonomous agents to query and trade against.

WHY THIS *matters*

Prediction markets only matter if the people funding them survive.

Omniverse makes providing liquidity survivable and makes borrowing liquidation-free — and the math that does it runs on-chain today, on Arbitrum Sepolia, not in a backend somewhere.

THE TEAM



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Blockchain Society · IIT Roorkee



OMNIVERSE — liquidity, defended.

THANK

you

omniverse-99so.vercel.app